Chapter 1. Introduction

Experiment, Design, and Statistics

1 Experiment, Design, and Statistics

- In any experiment, the experimenter is attempting to **draw certain inferences** or **make a decision** about some hypothesis.
- Decision-making needs **collecting data**.
- *Example (umbrella).* To decide whether or not to carry an umbrella on a given morning, one collects certain data: checking the barometer, looking at the sky, reading the newspaper forecast, listening to the radio, and so on. After quickly assimilating all the available data, a decision is made.
- **Statistics** is the science of making decisions (problem-solving) in the presence of variability (uncertainty).
- Decision-making based on statistical tools always involves the collection of data, so **how the data are collected** becomes extremely important.
- **Experimental designs** are used to help reduce the experimental error in data collection.
- **Randomization** is used to average out the effects of many (uncontrolled) variables in an experiment.

Definition 1.1: Basic principles of experimental design

(i) **Randomization.** Both the allocation of experimental material and the order in which the individual runs are performed are randomly determined. Statistical methods require that the observations (errors) be independently distributed random variables; randomization usually makes this assumption valid and "averages out" the effects of extraneous (lurking) factors.

(ii) **Replication.** An independent repeat run of each factor combination. Replication (a) allows estimation of the experimental error σ^2, which is the basic unit of measurement for deciding whether observed differences are real, and (b) gives a more precise estimate of a mean: if σ^2 is the variance of one observation, the sample mean of n replicates has variance σ^2/n.

(iii) **Blocking.** A technique to improve precision of comparisons among the factors of interest by reducing/eliminating the variability transmitted from nuisance factors. A block is a set of relatively homogeneous experimental conditions.

Remark 1.2: Why replication shrinks variance

If $y_1, \dots, y_n$ are independent with common variance σ^2, then

$$\operatorname{Var}(\bar{y}) = \operatorname{Var}\Big(\frac{1}{n}\sum_{j=1}^{n} y_j\Big) = \frac{1}{n^2}\operatorname{Var}\Big(\sum_{j=1}^{n} y_j\Big) = \frac{1}{n^2}\sum_{j=1}^{n}\operatorname{Var}(y_j) \quad \text{(independence)}$$
$$= \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

Hence, doubling the number of replicates halves the variance of a treatment mean. This simple fact is used repeatedly in every ANOVA derivation of later chapters (e.g. $\operatorname{Var}(\bar{\varepsilon}_{i\cdot}) = \sigma^2/n$).

2 Role of Statistics in Experimentation

- **Project planning phase**
  1. What is to be measured? (choice of the response variable)
  2. How large is the likely variation?
  3. What are the influential factors?
- **Experimental design phase**
  1. Control some known sources of variation (blocking).
  2. Allow estimation of the size of the uncontrolled variation (replication).
  3. Permit an investigation of suitable models.
- **Statistical analysis phase**
  1. Make inferences on design factors.
  2. Guide subsequent design.
  3. Suggest more appropriate models.

3 The General Workflow Used in This Course

Derivation

Every design studied in this course is analyzed with the same three steps (illustrated with the two-sample model $y_{ij} = \mu_i + \varepsilon_{ij}$, $\varepsilon_{ij} \sim N(0, \sigma^2)$, i.e. $y_{ij} \sim N(\mu_i, \sigma^2)$, $H_0 : \mu_1 = \mu_2$):

① **Estimation**: fit the model, $y_{ij} = \hat{\mu}_i + e_{ij}$ (fitted value + residual);

② **Inference**: test hypotheses / build confidence intervals on the parameters;

③ **Residual analysis**: check the model assumptions (normality, constant variance, independence) using the residuals e_{ij}.

Design	Model	Chapter
Two-sample / paired comparison	$y_{ij} = \mu_i + \varepsilon_{ij},\ \ y_{ij} = \mu_i + \beta_j + \varepsilon_{ij}$	2
Completely randomized design (CRD)	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	3
RCBD / BIBD	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	4
Latin / Graeco-Latin square	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$	4
Factorial design	$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	5
2^k factorial, blocking & confounding	effects via contrasts	6, 7
Regression approach	$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$	10
Random / mixed effects	$\tau_i \sim N(0, \sigma_\tau^2)$	13
Nested and split-plot	$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}, \ldots$	14
Analysis of covariance	$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$	15

4 Selected exercises

Motivation. The three principles above can be justified by short calculations: randomization makes the comparison unbiased whatever the units are like, it neutralizes a hidden time trend, and only genuine replicates give the σ^2/n variance.

Exercise 1.1 (Randomization makes the comparison unbiased)

$N = n + m$ experimental units have fixed unit effects $u_1, \dots, u_N$ (unknown). A completely randomized design assigns a simple random sample A of n units to treatment 1 and the other m units (B) to treatment 2, and unit k responds $y_k = u_k + \tau_1$ or $y_k = u_k + \tau_2$ (additivity). With $\bar{u} = \frac{1}{N}\sum_k u_k$ and $S_u^2 = \frac{1}{N-1}\sum_k (u_k - \bar{u})^2$, show that, over the randomization,

$$\mathrm{E}(\bar{y}_A - \bar{y}_B) = \tau_1 - \tau_2, \qquad \mathrm{Var}(\bar{y}_A - \bar{y}_B) = S_u^2\Big(\frac{1}{n} + \frac{1}{m}\Big).$$

Solution [Supplement].

Step 1 (reduction). $\bar{y}_A - \bar{y}_B = \tau_1 - \tau_2 + \bar{u}_A - \bar{u}_B$, and since $n\bar{u}_A + m\bar{u}_B = N\bar{u}$,

$$\bar{u}_A - \bar{u}_B = \bar{u}_A - \frac{N\bar{u} - n\bar{u}_A}{m} = \frac{N}{m}(\bar{u}_A - \bar{u}).$$

Step 2 (indicators). Let $I_k = \mathbf{1}\{k \in A\}$. Every unit is equally likely to be chosen, so $\mathrm{E}I_k = n/N$, and for $k \neq l$, $\mathrm{E}(I_k I_l) = \frac{n(n-1)}{N(N-1)}$, hence

$$\mathrm{Var} I_k = \frac{n}{N}\Big(1 - \frac{n}{N}\Big), \qquad \mathrm{Cov}(I_k, I_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3 (mean). $\bar{u}_A = \frac{1}{n}\sum_k I_k u_k$ gives $\mathrm{E}\bar{u}_A = \frac{1}{n}\sum_k \frac{n}{N} u_k = \bar{u}$, so $\mathrm{E}(\bar{y}_A - \bar{y}_B) = \tau_1 - \tau_2$.

Step 4 (variance). Write $v_k = u_k - \bar{u}$, so $\sum_k v_k = 0$ and $\bar{u}_A - \bar{u} = \frac{1}{n}\sum_k I_k v_k$. Using $\sum_{k \neq l} v_k v_l = (\sum_k v_k)^2 - \sum_k v_k^2 = -\sum_k v_k^2$,

$$\begin{aligned}\mathrm{Var}\Big(\sum_k I_k v_k\Big) &= \frac{n(N-n)}{N^2}\sum_k v_k^2 + \frac{n(N-n)}{N^2(N-1)}\sum_k v_k^2 \\ &= \frac{n(N-n)}{N(N-1)}\sum_k v_k^2 = \frac{n(N-n)}{N} S_u^2.\end{aligned}$$

Hence $\mathrm{Var}(\bar{u}_A) = \frac{N-n}{nN} S_u^2 = \frac{m}{nN} S_u^2$ and, by Step 1,

$$\mathrm{Var}(\bar{y}_A - \bar{y}_B) = \frac{N^2}{m^2} \cdot \frac{m}{nN} S_u^2 = \frac{N}{nm} S_u^2 = S_u^2\Big(\frac{1}{n} + \frac{1}{m}\Big).$$

No distributional assumption on the units was used: the random assignment itself creates the "error" and makes it average out. S_u^2 plays the role of σ^2 in the two-sample formula of Ch. 2. ■

Exercise 1.2 (A hidden time trend)

Two treatments are each run n times in $N = 2n$ consecutive runs, and the response drifts linearly with run order k: $y_k = \gamma k + \tau_{(k)} + \varepsilon_k$ (γ unknown, $\tau_{(k)}$ the treatment of run k, ε_k iid with mean 0).
(a) If all treatment-1 runs come first, find the bias of $\bar{y}_1 - \bar{y}_2$ for $\tau_1 - \tau_2$.

(b) If the run order is randomized, show the estimator is unbiased and find the variance contributed by the trend.

Solution [Supplement].

(a) The average run index is $(n+1)/2$ for treatment 1 and $n+(n+1)/2$ for treatment 2, so $\mathrm{E}(\bar{y}_1-\bar{y}_2)=\tau_1-\tau_2-\gamma n$: the bias $-\gamma n$ grows with the experiment and cannot be separated from the treatment difference.

(b) Apply Exercise 1.1 with unit effects $u_k=\gamma k$ (the errors average to zero independently of the assignment). The estimator is unbiased, and with $\sum_{k=1}^{N}(k-\frac{N+1}{2})^2=\frac{N(N^2-1)}{12}$,

$$S_u^2=\gamma^2\frac{N(N+1)}{12},\qquad \mathrm{Var}_{\mathrm{trend}}=S_u^2\cdot\frac{2}{n}=\gamma^2\frac{2n(2n+1)}{12}\cdot\frac{2}{n}=\gamma^2\frac{2n+1}{3}.$$

Randomization turns a systematic bias into extra random error; blocking on time would remove even that. ■

Exercise 1.3 (Replicates versus repeated measurements)

Suppose $y_1,\dots,y_n$ each have variance σ^2, but every pair has correlation $\rho\geq 0$ (e.g. n measurements on one experimental unit instead of n units).
(a) Show $\mathrm{Var}(\bar{y})=\frac{\sigma^2}{n}\{1+(n-1)\rho\}$ and that it tends to $\rho\sigma^2$, not to 0, as $n\to\infty$.
(b) Show that $\mathrm{E}(S^2)=(1-\rho)\sigma^2$, so the sample variance underestimates the variability that matters for comparing units.

Solution [Supplement].

(a) $\mathrm{Var}(\sum_j y_j)=\sum_j\mathrm{Var}(y_j)+\sum_{j\neq l}\mathrm{Cov}(y_j,y_l)=n\sigma^2+n(n-1)\rho\sigma^2$; divide by n^2. The limit is $\rho\sigma^2>0$ when $\rho>0$.

(b) $\mathrm{E}\sum_j(y_j-\bar{y})^2=\sum_j\mathrm{E}(y_j-\mu)^2-n\mathrm{E}(\bar{y}-\mu)^2=n\sigma^2-\sigma^2\{1+(n-1)\rho\}=(n-1)(1-\rho)\sigma^2$.

Thus repeated measurements both shrink the apparent error and stop reducing $\mathrm{Var}(\bar{y})$: only independent replicates (the remark above uses independence) deliver σ^2/n and an honest estimate of σ^2. ■